Project Name: ConnectHub – Student Networking
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Design

Introduction

The design phase translates the functional and technical requirements into a blueprint for the system's architecture, modules, and user interactions. It ensures that the system components are organized efficiently and can be developed, tested, and maintained effectively. For ConnectHub—a LinkedIn-style academic collaboration platform—the design approach follows a modular structure using UML diagrams and a microservices-based architecture, enabling scalability, flexibility, and clean separation of concerns.

UML Diagrams

Unified Modeling Language (UML) diagrams are used to visualize the structure and behavior of the system. The following diagrams illustrate how ConnectHub works internally and how different components interact.

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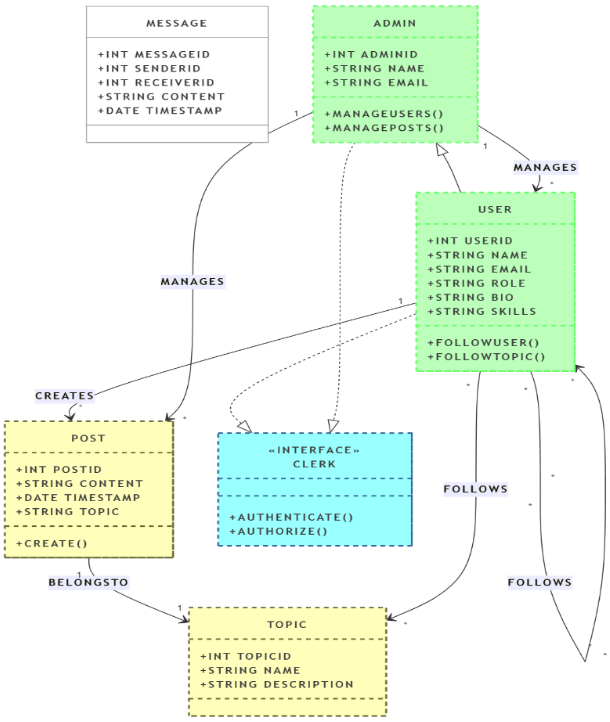
The UML diagram for ConnectHub illustrates the structural and behavioral design of the system. The **class diagram** defines key entities such as User, Post, Topic, Message, and Admin, each with specific attributes and methods. Relationships show that users can create posts, follow topics, and send messages, while admins manage users and content.

The **activity diagram** outlines the user workflow, from login to interacting with posts, topics, and messages. The **use case diagram** highlights system functionalities available to both users and admins.

Together, these diagrams provide a comprehensive view of how ConnectHub operates and how its components interact.

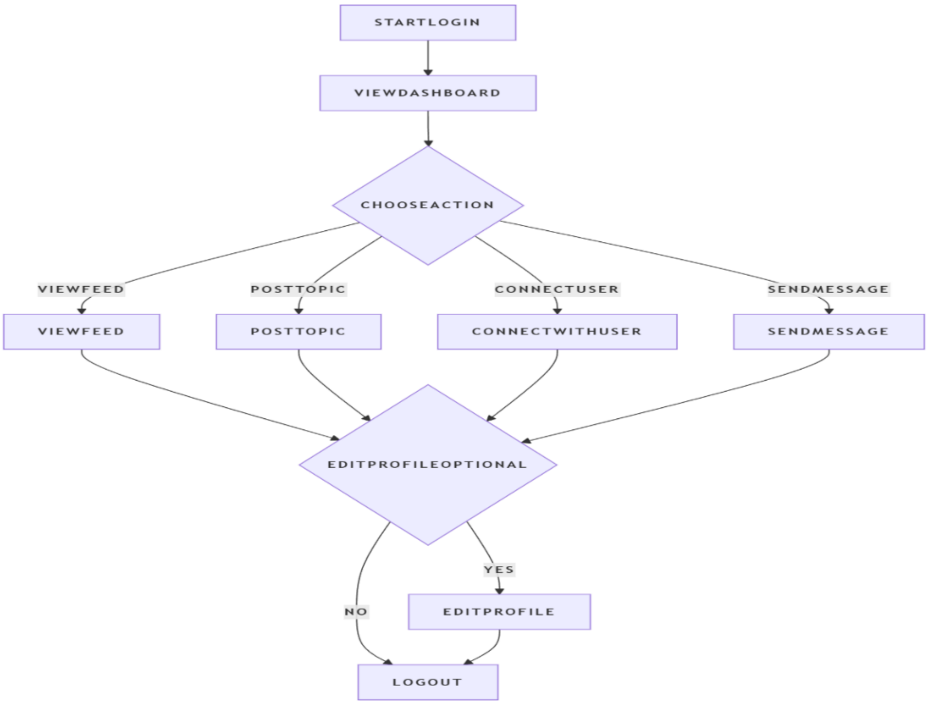
Class Diagram

The class diagram represents the static structure of the ConnectHub system by showing classes, attributes, methods, and relationships between entities.



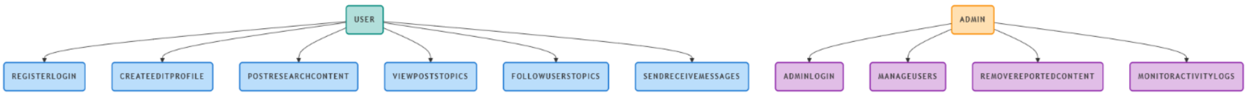
Activity Diagram

The activity diagram describes the dynamic flow of activities in the ConnectHub application from a user perspective.



Use Case Diagram

The use case diagram shows the interaction between actors (users and admin) and system functionalities.



Module Design and Organization

ConnectHub is designed using a modular microservices architecture. Each module is independent and responsible for specific functionality, making it easier to scale, maintain, and deploy.

Microservice Architecture Diagram